

MSDA9215: Big Data Analytics

April/May 2025

Hands-on Neo4j: US Road Network

Temitope Oguntade

Hands-on(NeoJ): US Road Network

You are given a dataset of US road network with the description below:

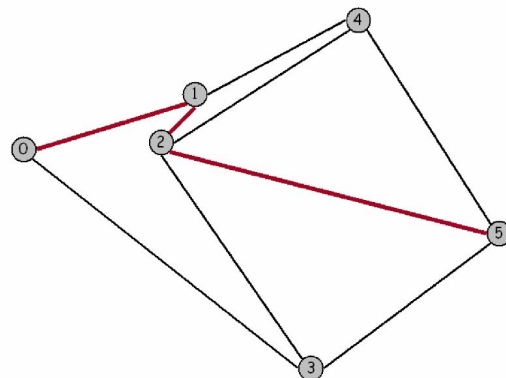
Maps. For this project we will be working with maps, or graphs whose vertices are points in the plane and are connected by edges whose weights are Euclidean distances. Think of the vertices as cities and the edges as roads connected to them.

To represent a map in a file, we list the number of vertices and edges, then list the vertices (index followed by its x and y coordinates), then list the edges (pairs of vertices), and finally the source and sink vertices. See the format to the right - - ->>>.

The full dataset contains **87,575** intersections and **121,961** roads in the continental United States. The graph is very sparse - the average degree is 2.8.

```
6 9
0 1000 2400
1 2800 3000
2 2400 2500
3 4000 0
4 4500 3800
5 6000 1500
```

```
0 1
0 3
1 2
1 4
2 4
2 3
2 5
3 5
4 5
```



Task:

As big data analysts, do the following tasks and submit a report for a 10 minute presentation.

Deadline: 2 weeks.

Group Composition: Max. of 3 students per group. Cannot have same members of Project 1 in the same group.

All group members will be required to present and can be selected at random to represent the group.

1. Query to find the total number of intersections and roads:

2. Find the shortest path between two intersections:

3. Query to find all intersections with a degree greater than a specific threshold (e.g., $\text{degree} > 3$):

4. Analyze the centrality of intersections using the Betweenness Centrality algorithm:

Betweenness centrality is a key concept in network analysis that helps identify the most critical nodes (or intersections in the context of a road network) within a graph. A node with high betweenness centrality lies on many of the shortest paths between other nodes. In other words, it acts as a bridge or bottleneck that connects different parts of the network.

5. Create a simple dashboard using Plotly (or other options) to display key metrics:

Total Intersections. Total Roads. Ideas?

6. Degree Distribution Bar Chart:

The distribution of node degrees (i.e., how many roads connect to each intersection). This will help visualize how connected the intersections are, showing whether most intersections have a small number of connections or if there are some highly connected hubs.

7. **Top 10 Most Connected Intersections:**

The top 10 intersections with the highest degree (most connections). Identifies the most critical intersections in terms of connectivity.

8. **Intersection Categories by Degree:**

Categorize intersections by their degree (e.g., low, medium, high connectivity) and show the number of intersections in each category. Provides a quick understanding of how intersections are distributed based on their connectivity levels.

9. **Degree Distribution Bar Chart:**

The distribution of node degrees (i.e., how many roads connect to each intersection). This will help visualize how connected the intersections are, showing whether most intersections have a small number of connections or if there are some highly connected hubs.